W-L-00311B(GSA-FSS)
March 1, 1969
INTERIM REVISION OF
Fed. Spec. W-L-311a
January 14, 1959

INTERIM FEDERAL SPECIFICATION

LIGHT, DESK, INCANDESCENT, FLEXIBLE COLUMN

This Interim Federal Specification was developed by General Services Administration, Federal Supply Service, Washington, D. C. 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

The General Services Administration has authorized the use of this Interim Federal Specification as a valid except to Federal Specification W-L-311a, dated January 14, 1959.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers flexible-arm, incandescent, electric desk lights, with weighted base or screw-clamp base.

1.2 Classification.

1.2.1 Types and classes. Desk lights covered by this specification shall be of the following types and classes (see 6.2):

Type I - Lights, desk, weighted base.
Class 1 - Light, desk, weighted base with parabolic-shaped reflector.
Class 2 - Light, desk, weighted base with trough-shaped reflector.

Type II - Lights, desk, screw-clamp base.
Class 1 - Light, desk screw-clamp base with parabolic-shaped reflector.
Class 2 - Light, desk, screw-clamp base with trough-shaped reflector.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

J-C-175 - Cable Assembly, Power, Electrical, (3-Wire, 3-Prong, Grounding Plug Connector, for 125-volt equipment).
J-C-580 - Cord, Flexible, and Wire, Fixture, (Electrical 0-to-600-volt service).
W-C-596 - Connector, Plug, Electrical; Connector, Receptacle, Electrical.
W-L-101 - Lamps, Electric, Incandescent, Large, Tungsten-Filament.
W-L-142 - Lampholders (Sockets); Medium-Screw-Shell, General Service.
W-S-610 - Splice, Conductor.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleater-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Fiber.
PPP-T-60 - Tape; Pressure Sensitive Adhesive. Waterproof-for Packaging and Sealing.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 595 - Colors.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.)

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-P-116 - Preservation, Methods of.
MIL-B-138 - Boxes, Wood, Fiberboard-Lined for Overseas Shipment (For Weight of Contents not Exceeding 500 pounds).
MIL-B-10377 - Box, Wood, Cleated, Veneer, Paper Overlaid.
MIL-T-704 - Treatment and Painting of Materiel.
MIL-D-1000 - Drawings, Engineering and Associated Lists.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U. S. Military Property.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Underwriters' Laboratories, Inc., Standards (UL):

UL-153 - Portable Electric Lamps.

(Application for copies should be addressed to Underwriters' Laboratories, Inc., Walt Whitman Road, Melville, New York 11749; 207 East Ohio Street, Chicago, Illinois 60611; or 1655 Scott Boulevard, Santa Clara, California 95050.)

3. REQUIREMENTS

3.1 Preproduction model. Unless otherwise specified (see 6.2), a preproduction model shall be submitted for examination, test, and inspection in accordance with section 4, and such other tests as are necessary to determine compliance with the provisions of this specification and the contract. In all respects not specifically covered by this specification, the preproduction model shall represent the identical quality of materials, workmanship, and design which will be used in the fabrication of the production units. Any fabrication of equipment to contractual requirements without approval of the preproduction model shall be at the contractor's risk. The approved preproduction model shall be used as a guide to inspect production units and may be presented for acceptance as the last unit of the contract. Examination and testing, whenever practicable, will be conducted at the contractor's plant, but the Government reserves the right to perform any of the inspections set forth in this specification at a laboratory of their choice (see 6.2).

3.2 Description. The lights shall have one incandescent lamp and shall consist of a reflector, lampholder, switch, flexible tubing, column, base, and electric cord. The two types of lights shall conform to UL 153 and MIL-STD-108, and as specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.

3.3 Fire and casualty hazards. Each contractor shall submit to the contracting agency proof that the light he proposes to supply under this specification conforms to the applicable requirements of UL 153 regarding fire and casualty hazards. The label or listing with reexamination of the Underwriters' Laboratories, Inc., will be accepted as evidence that the device conforms to this requirement.

3.3.1 In lieu of the label or listing with reexamination, the contractor may submit independent proof satisfactory to the contracting agency that his light conforms to the published standards, including methods of tests, of UL 153.

3.3.2 Compliance with the above preliminary requirements in regard to fire and casualty hazards does not absolve the contractor from complete compliance with the other requirements of this specification in order to secure acceptance of the material or appliance.

3.4 Design.

3.4.1 General. The type I light shall consist of a weighted base and an attached column (see paragraph 3.8) supporting at the upper end a lampholder and reflector. The type II light shall conform to the requirements of type I except it shall have a clamp-type base.

3.4.2 Electrical rating. Unless otherwise specified (see 6.2), the light shall operate from a 115/125-volt, 60 cycle, alternating-current source.

3.4.3 Size. With the column positioned vertically the light shall be not less than 24 inches nor more than 28 inches high when measured from the bottom of the base along the axis of the column to the farthest point on the rim of the reflector. Flexible tubing of the column shall be of sufficient length to permit at least 8 inches of flexibility.

3.4.4 Dielectric strength. Each light shall have a dielectric strength of 1,000 volts ac plus twice the rated operating voltage when tested in accordance with paragraph 4.4.2.

3.4.5 Material. The material shall be as specified herein. Material not definitely specified shall be of the quality regularly used by the manufacturer in the product he sells on the commercial market. The material shall be free from all defects and imperfection that might affect the serviceability and appearance of the finished product.

3.4.6 Devices. The light shall be furnished complete with reflector, lampholder, lamp, wiring and all other devices necessary for operation.

3.5 Construction. The light shall be constructed of metal. Bases, colums, and supports shall be fabricated from one or more of the following: iron, steel, brass, bronze, aluminum, aluminum alloy, zinc alloy, or magnesium alloy. White-metal castings(lead base) shall not be used in bases, columns, or supports. Each light shall consist of a metal base with one metal column or other support, on the upper end of which shall be mounted a reflector and reflector housing. The reflector may be integral with the housing or may be separate and enclosed by a housing. The light shall be sufficiently sturdy to withstand without damage the application of any force or stress to which it will be normally subjected in service and shipment.

3.5.1 Exposed edges. Exposed edges of sheet metal or other metal parts shall be finished so that personal injury due to sharp edges will not occur under any normal circumstances.

3.5.2 Wiring. The only exposed wiring shall be from the base to the plug connector and shall enter the light through or at the base (see 3.6). All other wiring shall be entirely concealed.

3.6 Electrical cord. Unless otherwise specified (see 6.2), each light shall be provided with an electrical cord having an exposed length of not less than 9 feet with molded-on or clampled-on, parallel-blade attachment plug. The cord shall be type SP-2 or SPT-2, in accordance with J-C-580, and the plug shall be in accordance with W-C-596. When specified (see 6.2), a 3 conductor cord shall be supplied and shall be in accordance with type I of J-C-175. The color of the cord shall be brown or gray as specified (see 6.2).

3.6.1 Splices. All splice connections shall be made with splice conductors in accordance with Class 1, style G of W-S-610.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.

3.7 Base.

3.7.1 Type I. The base shall have a covering of spun or shaped steel not less than 0.018 inch thick or of zinc alloy pressure-cast diecasting not less than 0.065 inch thick. The base area and weight shall be sufficient to cause the light to be free standing on a horizontal surface and to prevent the light from overturning when the base is inclined at an angle of 8 degrees from the horizontal in any direction as required by UL-153. A felt pad not less than 3/64 inch thick and not less than 50% wool, of a color to harmonize with the light finish, shall be cemented to the bottom of the base.

3.7.2 Type II. The base shall be of formed steel or cast iron and shall be equipped with steel mounting screws which will permit opening up to at least 2 inches. Felt as specified in 3.7.1 shall be permanently affixed to the two opposing binding surfaces of the clamp.

3.8 Column. The light column shall be of metal and shall consist of an upright attached to the base and a flexible arm attached to the upper end of the upright by a threaded adaptor, or it shall consist of a flexible arm attached to the base. The reflector housing shall be attached at the upper end of the column.

3.8.1 Movement of column. Starting with the light on a horizontal surface so that the front and back edges of the reflector housing are in a plane parallel to the surface on which the light rests, the column shall be sufficiently flexible to permit the reflector housing, lampholder and lamp to be moved vertically upward through an angle of at least 90 degrees. Starting as above, the column shall also permit the reflector housing, lampholder and lamp to be moved through a horizontal angle of at least 135 degrees.

3.9 Reflector housing assembly. The reflector housing and reflector shall be of metal and shall be securely fastened to the flexible tubing or the lampholder at its upper end by clamp, snap-on, or screw-on method. The reflector housing and reflector may be integral or consist of a separate housing and reflector. The size and construction shall permit the insertion and removal of a standard 60 watt medium screw base incandescent lamp conforming with W-L-101. When inserted into the socket, no portion of the lamp shall project beyond the plane of the reflector rim.

3.9.1 Class 1, parabolic-shaped reflector. The reflector shall be parabolic shaped and the plane bounded by the rim shall be essentially circular.

3.9.2 Class 2, trough-shaped reflector. The trough shaped reflector shall have an overall length and depth to accomodate the size lamp specified in 3.9 in a horizontal position inside the reflector. The rim of the reflector shall be on a single plane.

3.10 Lampholder. The lampholder shall permit easy insertion and removal of the lamp. Contacts shall be held securely in place and shall be insulated from one another. The lampholder shall conform to the applicable requirements of type I, class A or C of W-L-142, except the shell may be of brass or aluminum.

3.11 Performance.

3.11.1 Illumination. The light when equipped with a lamp shall provide the following minimum illumination on a horizontal surface when tested as specified in 4.4.1.

Distance in inches measured vertically between the center of the lamps to the illuminometer:	Minimum illumination in foot-candles indicated on the illuminometer
Inches	Foot-candles, minimum
12	75

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.

3.12 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-D-1000 shall govern the manufacturer's part numbers and changes thereto.

3.13 Finish. Unless otherwise specified (see 6.2), all exposed metal parts shall be cleaned, free of dirt, oil, grease and moisture, treated, and painted in accordance with MIL-T-704. Reflecting surfaces other than aluminum shall be white. All surfaces except aluminum reflecting surfaces, shall be baked enamel and the finish color shall conform to Fed. Std. No. 595 and shall be either smooth or crackle finish of brown, bronze, or gun metal finish as specified in the Invitation for Bids (see 6.2). All enamel surfaces shall be free from tears, star marks, crazing, blisters, hairlines, chipping, pinholes, and other irregularities or defects.

3.14 Nameplate. Unless otherwise specified in the Invitation for Bids (see 6.2), as indentification plate permanently and legibly providing the necessary information in accordance with MIL-STD-130 shall be securely attached to the light in an inconspicuous location.

3.15 Workmanship. The light shall be free from defects such as chips or discoloration in the paint which detract from the appearance or impair serviceability. Electrical connections shall be secure, metal joints and seams shall be tight, and all edges shall be rounded or beveled. The entire reflector and light assembly shall be firmly seated at all points and locked with bolts, nuts and washers wherever necessary.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsibile for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Inspection. Sampling for visual and dimensional inspection shall be conducted in accordance with MIL-STD-105, inspection level II. Acceptance shall be based on an acceptable quality level (AQL) of 2.5 defects per hundred units.

4.2.2 Tests. Sampling for tests (see 4.4.2 and 4.4.3). A sample of equipment shall be selected from each inspection lot in accordance with MIL-STD-105, inspection level S-2. Acceptance shall be based on an AQL of 1.5 defects per hundred units.

4.3 Inspection. Samples selected in accordance with 4.2 shall be inspected for compliance with the following requirements:

Electrical rating (see 3.4.2)
Exposed edges (see 3.5.1)
Wiring (see 3.5.2)
Electrical Cord (see 3.6)
Base (see 3.7)
Column (see 3.8)
Reflector housing assembly (see 3.9)
Joints and connections (see 3.8 and 3.9)
Lampholder (see 3.10)
Finish (see 3.13)
Nameplate (see 3.14)
Workmanship (see 3.15)

4.3.1 Flexibility examination. With the flexible column bent so that the light is at the maximum horizontal distance from the center line of the junction of the flexible column and the base, there shall be no tipping or instability. The unit shall be capable of passing the aforementioned test for any radial position of the flexible column.

4.3.2 Dimensional examination. Any dimension that is not within the specified tolerance shall be classified a defect.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.

4.4 Tests.

4.4.1 Photometric. The sample light shall be equipped for test with a 60-watt. incandescent lamp which has been aged not less than 10 hours and which has not had more than 50 hours of use. A sample light shall be placed on a horizontal surface so that the front and back edges of the reflector housing are in a plane parallel to the horizontal surface on which the light rests and at a vertical distance of 12 inches from the horizontal surface or at the greatest possible vertical distance less than 12 inches which satisifes these conditions. An illuminometer or other suitable instrument capable of being read in foot-candles within the range of illumination values given in 3.11.1 and which has been calibrated to indicate the values of illumination of incandescent light shall be used. The readings shall be taken perpendicular to the normal of the lamp and, if a cosine corrected meter is not used, shall be multiplied by the cosine of the angle between the vertical and the direction of the normal to the lamps to obtain the illumination on the horizontal surface. It is often more convenient to measure the angle between the normal to the lamps and the horizontal; if this angle is used, the readings shall be multiplied by the sine of the angle in order to obtain the horizontal reading. Lamps providing less than the minimum illumination in foot-candles specified in 3.11.1 to be indicated on the illuminometer shall constitute failure of this test. This test is conducted on the preproduction model only.

4.4.2 Dielectric strength. Lampholders shall withstand the application for one minute of an alternating current having an effective voltage of 1,000 volts plus-twice the rated voltage of the lamp. The test shall be made between terminals of opposite polarity, between current-carrying parts and grounded metal, or between current-carrying parts and metal parts exposed on the surface of the mounting. Arcing or shorting of electrical parts shall constitute failure of this test.

4.4.3 Operation. Testing shall be performed on a sample desk light, equipped with a standard 60 watt, medium, screw-base, incandescent lamp connected to a 115 volt, 60-cycle, a-c source. The switch shall be operated to light the lamps. The reflector housing shall be moved to determine compliance with the requirements of paragraph 3.9. Inability of the switch to turn the lamp on or off, inability to move the reflector housing freely in any of the positions stated shall be classified as a defect.

4.5 Inspection of preparation for delivery. An inspection shall be made to determine that the packaging, packing, and marking comply with section 5 of this specification. Defects shall be scored in accordance with table I. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery selected just prior to the closing operations. Samples shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

TABLE I. Classification of preparation for delivery defects.

Examine	Defects
Markings (Exterior and Interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing, or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Bulging or distortion of container.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. Each light shall be packaged in accordance with MIL-P-116, method IC-5, utilizing containers conforming to W5C of PPP-B-636. The containers shall be closed with adhesive and stitching as specified in the appendix to the box specification adn in addition, all corners and joints shall be waterproof sealed with 2 inch minimum width tape conforming to PPP-T-60.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.

5.1.2 Level B. Each light shall be packaged in a fiberboard box conforming to PPP-B-636, class domestic. Bursting strength of the fiberboard shall be not less than 200 p.s.i. Blocking and bracing shall be provided within the container to prevent movement of the light. Closure shall be in accordance with the appendix to PPP-B-636.

5.1.3 Level C. The lights shall be packaged in accordance with the suppliers commercial practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Twelve packages (12 lights of one type and class only, packaged as specified in 5.1) shall be packed in the most compact manner in a snug-fitting shipping container constructed and strapped to conform to V3C or V3s of PPP-B-636. Alternatively, shipping containers constructed, closed, and strapped to conform to overseas type of PPP-B-591, PPP-B-601, or MIL-B-10377; or class 2, style 2 or 4 of PPP-B-621; or style 1, 2, or 2A, class 3 use of PPP-B-585; or MIL-B-138 may be used. Only one type, style, and size shipping container shall be used for any one item on any one contract.

5.2.2 Level B. Six unit packages, of one stock number, shall be packed in a fiberboard box conforming to PPP-B-636, type, class, and style optional. Bursting strength of the fiberboard shall be not less than 275 p.s.i. Closure of the box shall be in accordance with the appendix to PPP-B-636.

5.2.3 Level C. Lights shall be packed to insure carrier acceptance and safe delivery to destination in containers complying with the rules and regulations applicable to the mode of transportion.

5.3 Standard Pack (Civil Agency Procurement). The Standard Pack for civil agency procurements shall be as specified in paragraphs 5.1.2 and 5.2.2 (see 6.2 and 6.4).

5.4 Marking.

5.4.1 Civil agencies. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.4.2 Military activities. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The lights are intended for use on desks, tables, drawing boards, benches machines, et cetera to shield the light from the worker's eyes and direct maximum intensity on the work area.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Type and class of light required (see 1.2).
(c) Preproduction model, if required and laboratory where preproduction inspection shall be conducted (3.1).
(d) Electrical rating if other than specified (see 3.4.2).
(e) Type of electrical cord and plug if other than specified (see 3.6).
(f) Cord color required (see 3.6).
(g) Length of column (see 3.4.3).
(h) Finish if other than specified and finish color required (see 3.13).
(i) Nameplate information, if other than specified (see 3.13).
(j) Responsibility for inspection, if other than specified (see 4.1).
(k) Selection of the applicable levels of packaging and packing (see 5.1, 5.2 and 5.3). Level B preservation and packaging is intended to provide economical but limited protection and should be specified only when it is determined the lights will be held in covered storage approximately one year from date of initial preservation and packaging.
(l) Quantity to be packed (see 5.2.2).
(m) Marking, if other than specified (see 5.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.

6.3 Standard pack for civil agencies. The standard pack requirements in 5.3 are intended for use in procurements of Stores Stock replenishments. Procuring officers should use the standard pack requirements when it is known that the material will be shipped from a supplier to a domestic warehouse, supply depot or intermediate storage point for temporary storage, subsequent issue and/or shipment to eventual user.

Preparing activity:

GSA - FSS

GSA DC 70-2541

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.

GENERAL SERVICES ADMINISTRATION - FEDERAL SUPPLY SERVICE

SPECIFICATION COMMENT SHEET

BUDGET BUREAU NO. 29-R0175

INSTRUCTIONS

This form provides a way for users of this specification to inform the originator of problems encountered in its use. It is not to be used to request changes to accommodate proprietary features. All comments will be considered and appreciated, but please do not expect a reply. To comment: detach, complete, fold, staple, and mail.

NOTE: Comments on this form do not constitute or imply authorization to waive any part of the document or serve to amend contractual requirements.

1. SPECIFICATION

W-L-00311B (GSA-FSS) Light, Desk, Incandescent, Flexible Column

2. CONTRACT NO. *(If any)*	3. QUANTITY ON CONTRACT *(Optional)*	4. DOLLAR VALUE *(Optional)*

5. GENERAL NATURE OF PROBLEM *(e.g., inspection difficulties, manufacturers unable to meet tolerances, containers collapse under normal warehousing conditions, etc.)*

6. SPECIFIC REQUIREMENTS AFFECTED *(Include paragraph number and lines of wording)*

7. SPECIFIC PROBLEMS *(e.g. tests in 4.2.2 will not assure that the battery will last required time; temperature ranges in table 2 do not conform to commercially available items.)*

8. RECOMMENDATIONS

9. NAME OF MANUFACTURER, ASSOCIATION, GOVT., AGENCY, ETC.	10. ADDRESS *(Number, Street, City, State and Zip Code)*

11. NAME AND TITLE OF SUBMITTER	12. DATE

FORM GSA FEB 69 2200

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.